Splitly — Project Vision and Scope

1. Document Purpose

This document defines the product vision, target users, current-state scope, future-state scope, product boundaries, assumptions, and expected outcomes for Splitly.

The document intentionally separates the current manual-entry workflow from the future AI-assisted receipt-scanning workflow. OCR-related code in the repository is treated as an experimental foundation rather than a completed current-state capability.

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2. Product Overview

Splitly is a responsive expense-sharing web application that helps a group record a shared bill, determine how much each participant owes, and track whether each participant has paid.

The current application supports manual bill creation with three splitting approaches:

1. **Equal split** — the bill is divided equally among selected participants.
2. **By-person split** — the creator specifies the exact amount owed by each participant.
3. **By-item split** — bill items are entered manually and assigned to one or more participants.

After a bill is created, the application provides a bill-detail view with participant-level amounts, payment progress, payment status, and reminder actions.

The future product state will add an AI-assisted receipt-scanning workflow that extracts bill data from a receipt image and pre-fills the existing bill-creation form. The user must review and confirm the extracted data before the bill is saved.

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3. Product Vision

> **To make shared dining expenses quick, fair, and transparent by reducing manual bill transcription while preserving user control over how every item and payment is allocated.**

Splitly is not positioned as a simple calculator that only divides a total by the number of participants. Its value comes from combining:

- structured bill records;
- flexible splitting methods;
- item-to-person allocation;

- payment tracking;
- a future AI-assisted data-entry flow;
- a mandatory human review step for financial accuracy.

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4. Mission Statement

> **Splitly enables groups to transform a physical receipt into a clear, reviewable, and trackable shared expense record, reducing calculation effort, payment confusion, and awkward follow-up conversations.**

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5. Product Positioning

5.1 Current Position

The current version is a **manual shared-bill management tool**.

Users enter bill details, select a payer and participants, choose a splitting method, save the bill, and track payment completion.

5.2 Future Position

The future version is an **AI-assisted shared-bill management tool**.

Users scan or upload a receipt, AI extracts structured data, the system pre-fills the bill form, and the user verifies the result before continuing with participant assignment and splitting.

5.3 Positioning Statement

For groups of friends who frequently share meals, coffee, and entertainment expenses, Splitly is a bill-sharing application that provides flexible allocation and payment tracking. Unlike using a calculator, notes application, or group chat, Splitly keeps bill data, participant obligations, and payment status in one structured workflow. Its future AI-assisted scanning feature will reduce repetitive manual entry without removing user review and control.

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6. Problem Statement

6.1 Primary Problem

When a group shares a restaurant or café bill, one person commonly has to:

- read the receipt;
- manually enter or recalculate the bill;
- identify which participant consumed each item;
- divide shared items;
- communicate each person's amount;
- remember who has already paid;
- remind unpaid participants.

Even when the arithmetic is simple, the full process is repetitive, error-prone, and socially inconvenient.

6.2 Current-State Pain Points

Pain point	Consequence
Manual transcription of bill information	Slow bill creation and risk of input errors
Re-entering every item for item-based splitting	High effort for long receipts
Switching between receipt, calculator, notes, and chat	Fragmented workflow
Unclear payment status	Repeated checking and follow-up
Manual reminders	Social discomfort for the payer
Lack of a shared record	Disagreement about amounts or payment status

6.3 Future-State Opportunity

AI-assisted receipt scanning can reduce the most repetitive step: converting an unstructured receipt image into structured bill data.

The AI feature does not replace the user. It produces a draft that must be reviewed because receipt images may be blurred, incomplete, folded, or formatted inconsistently.

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7. Target Users

7.1 Primary Persona — Social Dining Group

Persona name: Minh — Group Bill Organizer
Segment: Friends or young professionals sharing meals, coffee, and entertainment expenses
Typical age range: 18–35
Role in the workflow: The person who pays upfront or volunteers to create the bill

Characteristics

- Frequently joins meals, cafés, parties, movies, or entertainment activities with friends.
- Usually uses a smartphone during or immediately after the activity.
- Wants to complete the bill split quickly before the group disperses.
- May need equal, by-person, or item-based splitting depending on the situation.
- Wants a clear record of who owes what and who has paid.

Goals

- Create a shared bill with minimal effort.
- Allocate costs fairly.
- Avoid manual recalculation.
- Avoid sending repeated individual messages.
- Confirm that the collected amount matches the bill.

Frustrations

- Long receipts with many items.
- Shared items consumed by multiple people.
- Participants leaving before the split is complete.
- Needing to check bank transfers manually.
- OCR errors that are difficult to notice.

7.2 Secondary Persona — Travel Group Organizer

A person coordinating food, transport, accommodation, and activity expenses during a group trip.

Key additional needs include rapid entry of many bills and a clear history. Multi-currency and final trip-wide settlement are not included in the approved TV3 scope unless separately approved by the team.

7.3 Secondary Persona — Roommate Expense Coordinator

A roommate recording shared groceries, household supplies, utilities, or occasional shared purchases.

Recurring billing is not part of the approved current/future workflow in this document.

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8. Product Goals

8.1 Current-State Goals

- Provide a clear manual workflow for creating a bill.
- Support equal, by-person, and by-item splitting.
- Allow a bill item to be assigned to one or more participants.
- Record one designated payer for each bill.
- Display each participant's amount and payment status.
- Preserve bill history for later review.

8.2 Future-State Goals

- Allow users to upload or capture a receipt.
- Use AI to extract structured bill information.
- Pre-fill the existing bill-creation form.
- Make uncertain or missing data visible to the user.
- Require user review before saving.
- Preserve manual correction and manual-entry fallback.
- Reduce bill-creation effort without weakening financial accuracy.

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9. Current-State Scope

The current state is the manual bill-entry workflow that the team considers stable enough to demonstrate.

9.1 In Scope

- User authentication and protected access.
- Manual bill creation.
- Bill name.
- Category.
- Notes.
- Creation date.
- Payment deadline.
- One designated payer.
- Participant selection.
- Equal split.
- By-person split.

- By-item split.
- Item quantity and amount entry.
- Assignment of an item to one or more participants.
- Calculation and validation of participant amounts.
- Bill saving.
- Bill history and detail.
- Participant payment status.
- Payment progress.
- Payment reminders.

9.2 Current-State Limitations

- Bill data must be typed manually.
- Only one payer is stored for a bill.
- OCR is not accepted as a completed current-state feature.
- Explicit tax, service fee, tip, and discount fields are not confirmed in the current manual form.
- The repository documentation contains broader feature claims than the verified current workflow.
- AI payer selection, group-wide settlement optimization, recurring bills, and multi-currency are not part of the current state.

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10. Future-State Scope

The approved future state focuses on AI receipt scanning and automatic bill-form population.

10.1 In Scope

- Capture or upload a receipt image.
- Supported image validation.
- Receipt preview.
- AI/OCR processing state.

- Extraction of available receipt fields:
  - merchant or bill title;
  - transaction date;
  - total amount;
  - item names;
  - item quantities where available;
  - item prices;
  - category suggestion where feasible.
- Pre-filling the manual bill-entry form.
- User review and correction of extracted information.
- Selection of payer and participants.
- Selection of split method.
- Item assignment.
- Total and allocation validation.
- Saving the confirmed bill.
- Payment tracking after creation.
- Manual-entry fallback.
- Error message and retry flow.

10.2 Conditionally In Scope

The following fields may be included only when TV4 and TV5 confirm requirements and feasibility:

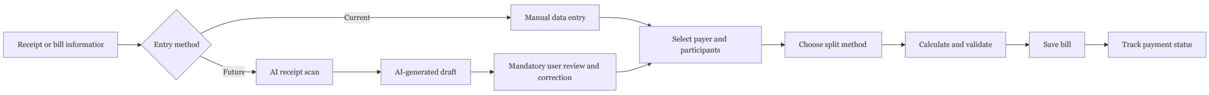
- tax;
- service charge;
- tip;
- discount or voucher;
- OCR confidence indicator;
- field-level warning state;
- receipt image retention policy.

10.3 Out of Scope for This Future-State Definition

- Fully autonomous saving without human review.
- Automatic bank transfer initiation.
- Multiple payers for a single bill.
- Cross-bill debt simplification.
- AI payer rotation.

- Group fund or wallet.
- Multi-currency conversion.
- Recurring bills.
- Advanced analytics.
- Dispute-resolution workflow.
- Native mobile application.
- Automatic interpretation of handwritten receipts unless separately validated.

11. Scope Boundary Diagram



12. Assumptions

- The primary use case involves a single receipt for one social dining event.
- A single user creates the bill.
- One person is designated as the payer in the current data model.
- Participants are identifiable through existing users or supported participant-selection mechanisms.
- The user has access to the physical or digital receipt.
- AI output is treated as a draft, not as authoritative financial data.
- The user is responsible for confirming the final total and allocation.
- Manual bill creation remains available after OCR is introduced.

13. Constraints

- Receipt quality varies significantly.
- Vietnamese receipt layouts are not standardized.
- Item names may be abbreviated.
- Totals may include tax, service charges, discounts, or rounding that are not represented as normal items.
- OCR/AI provider cost or usage limits may restrict testing.
- The current UI and data model may require changes to represent extraction confidence and receipt metadata.
- Financial calculations require deterministic validation independent of AI output.

- The future workflow must fit the team's approved schedule and available technical resources.

14. Dependencies

Dependency	Required from
Validated problem evidence and competitor gap	TV2
Product backlog and acceptance criteria	TV4
OCR provider, extraction schema, and technical feasibility	TV5
Schedule, cost, resource allocation, and go/no-go checkpoint	TV6
Approved stakeholders and project objectives	TV1
Figma Make prototype screens	UI/UX owner / TV3

15. Success Definition

The TV3 product workflow is considered successful when:

1. The distinction between current manual entry and future AI entry is unambiguous.
2. Every future AI step has a manual correction or fallback path.
3. The prototype demonstrates the complete primary persona journey.
4. Workflow steps map to features and future acceptance criteria.
5. No document presents experimental OCR as a completed current-state capability.
6. The bill cannot be saved when required financial data is invalid.
7. The final bill allocation reconciles with the confirmed bill total.

16. Repository Alignment Notes

- The current bill-creation code explicitly labels the flow as manual entry.
- The current code supports three splitting methods: equal, by person, and by item.
- The bill submission payload contains a single `payerId`.
- The OCR page currently contains an upload → AI recognition → bill creation concept and passes parsed data into the bill-creation page.
- For project documentation, OCR remains classified as future until the team validates its reliability and acceptance criteria.

- Repository documents use both “Spitly” and “Splitly”; all new submission documents should use **Splitly**.